Title and Objective

Neural Network Option Pricing

Approximating Black-Scholes European Call Prices with Feed-Forward Neural Networks

Objective

Evaluate whether a feed-forward neural network can learn the Black-Scholes European call pricing function and act as an efficient pricing surrogate.

Project group

Machine Learning for Finance

Research Question

Research question

Can a feed-forward neural network accurately approximate the Black-Scholes pricing function for European call options?

The project compares:

- the analytical Black-Scholes formula;
- Monte Carlo simulation;
- a supervised neural network pricing surrogate.

Black-Scholes SDE

The underlying asset follows the geometric Brownian motion

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t,$$

where r is the risk-free rate, σ is volatility and W_t is a Brownian motion.

The analytical European call price is used as the ground truth:

$$S_T = S_0 \exp\left((r - \tfrac{1}{2}\sigma^2)\,T + \sigma W_T\right).$$

European Call Option Pricing

A European call option has payoff

$$\Phi(S_T) = \max(S_T - K, 0).$$

Under the Black-Scholes model, its analytical price is

$$C_{\mathrm{BS}} = S_0 N(d_1) - K e^{-rT} N(d_2),$$

with

$$d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

Synthetic Dataset

Synthetic observations are sampled over the input space:

$$x = (S_0, K, T, r, \sigma), \qquad y = C_{\mathrm{BS}}(S_0, K, T, r, \sigma).$$

The final model also includes moneyness S_0/K as an engineered feature.

Variable	Range
Initial price S_0	$[50, 150]$
Strike K	$[50, 150]$
Maturity T	$[0.1, 2]$
Risk-free rate r	$[0, 0.05]$
Volatility σ	$[0.1, 0.6]$

Neural Network as Pricing Surrogate

The neural network learns the pricing map

$$f_\theta : (S_0, K, T, r, \sigma, S_0/K) \mapsto C_{\mathrm{BS}}.$$

Final architecture:

- fully connected feed-forward network;
- moneyness S_0/K included as engineered feature;
- SiLU hidden activations;
- mean squared error loss;
- Adam optimizer;
- train/validation/test split with feature scaling.

Experimental Pipeline

1. Sample synthetic contracts.
2. Compute Black-Scholes analytical prices.
3. Train the neural network on the supervised pricing task.
4. Estimate selected prices with Monte Carlo simulation.
5. Compare errors and analyse where approximation is harder.

Evaluation Metrics

The main quantitative metrics are:

- Mean Absolute Error;
- Root Mean Squared Error;
- coefficient of determination R^2;
- relative error diagnostics where meaningful.

Error analysis is performed against moneyness S_0/K, maturity T, and volatility σ.

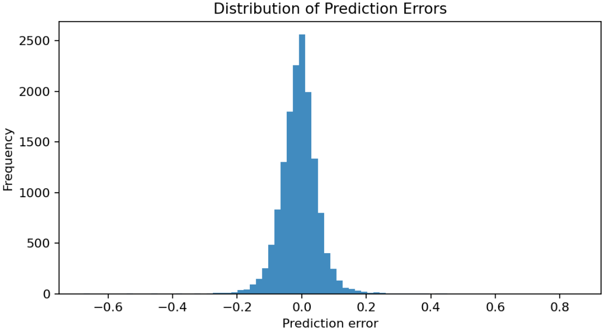
Main Results

Final experiment:

- 100,000 synthetic option contracts;
- neural network trained up to 200 epochs with early stopping;
- Monte Carlo benchmark with 50,000 paths on 512 test options.

Method	MAE	RMSE	R^2
Neural network	0.04290	0.06208	0.999993
Monte Carlo	0.08882	0.15790	0.999951

Error Analysis



Errors are concentrated near zero, supporting the low MAE and RMSE values.

Monte Carlo Benchmark

Simulated terminal prices are generated as

$$S_T = S_0 \exp\left((r - \tfrac{1}{2}\sigma^2)\, T + \sigma \sqrt{T} Z \right), \qquad Z \sim \mathcal{N}(0,1).$$

The discounted estimator is

$$\widehat{C}_{\mathrm{MC}} = e^{-rT} \frac{1}{M} \sum_{m=1}^{M} \max(S_T^{(m)} - K, 0).$$

- Evaluated on 512 deterministic test options.
- Provides a classical numerical benchmark with sampling error.
- The trained neural network gives deterministic forward-pass predictions.

Runtime Benchmark

Method	Time	Options/s
Black-Scholes	0.00091 s	1,101,095
Neural inference	0.03190 s	31,351
Monte Carlo	1.28235 s	780

Surrogate pricing point

After training, neural inference was about $40\times$ faster than Monte Carlo for pricing 1,000 options in the benchmark.

Additional ML Baseline: SVR

A reduced-scale Support Vector Regression benchmark was added as a classical ML reference.

Metric	Mean	Std
MAE	0.15088	0.00732
RMSE	0.23100	0.00592
R^2	0.999905	0.000005
MAPE	1.8253%	0.2422%

Evaluated on 5,000 synthetic samples over three seeds. SVR is useful as a classical baseline, while the neural network remains the main scalable surrogate model.

Discussion

- The network accurately learns the Black-Scholes pricing map in the sampled domain.
- Monte Carlo is also accurate, but remains simulation-based and stochastic.
- The SVR baseline gives an additional non-neural comparison on reduced datasets.
- The neural network is best interpreted as a pricing surrogate after training.
- The experiment validates the controlled Black-Scholes setting before moving to richer dynamics.

Limitations

- The project uses synthetic data generated under Black-Scholes assumptions.
- The model prices European calls only.
- The target function is known analytically, so the neural network is evaluated as a surrogate rather than as a replacement for financial theory.
- Generalization outside the sampled parameter ranges is not guaranteed.

Conclusions and Future Work

The research question is answered positively in the controlled Black-Scholes setting.

Main takeaway

A feed-forward neural network can approximate European call prices generated by the Black-Scholes formula with very high test-set accuracy.

Future extensions may include stochastic volatility, Heston dynamics, American options and Greeks estimation.